AI-Based Electricity Demand Intelligence System For Sri Lanka

Project ID
R26-IT4010

Project Proposal Report

DEWMINI D.M.D.P. – IT22630216

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

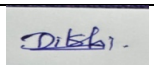
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Dewmini D.M.D.P.	IT22630216	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Nathalie de silva	20/03/2026	

ABSTRACT

Electricity demand management presents an ongoing problem for planners due to population increases, industrial development, and variations in weather patterns. Temperature, seasonality, and human behaviour all influence electricity consumption. Demand forecasting models primarily depend on historical data only and tend not to consider additional influences or possible outcomes.

Currently, demand forecasting systems lack scenario-based analyses, which would allow planners to explore the effects of extreme weather, public holidays, etc., on the demand for electricity. If planners cannot employ scenario-based analysis, they will likely struggle to anticipate the impact of sudden fluctuations in demand.

To remedy this issue, the present study proposes to develop a scenario-based decision support system that enables electricity planners to explore multiple future scenarios, with respect to demand and the resultant impacts of those scenarios. This system will incorporate machine learning techniques, Python for data analysis, and a web-based dashboard built with ReactJS/NodeJS to provide a visualization of results.

The resulting product will be a decision support tool to enhance electricity demand planning and management capabilities of electricity providers and improve energy distribution and operational efficiencies.

Keywords: Electricity Demand, Decision Support System, Scenario Analysis, Machine Learning, Energy Management.

ACKNOWLEDGEMENT

I would like to take this opportunity to thank my project supervisor, Dr. Nathali Silva, for her invaluable assistance, consistent support, and constructive criticism throughout the entire process of completing this research project. Her expertise and guidance assisted us significantly in comprehending the research problem and improving the quality of our work.

I would also like to thank the Department of Information Technology and the Sri Lanka Institute of Information Technology (SLIIT) for their provision of the academic setting, educational materials, and technical support that allowed me successfully to conduct this research project.

Finally, I would like to thank my family members and friends for their ongoing encouragement, moral support, and understanding while I finished the project.

Table of Contents

DECLARATION 2

ABSTRACT 3

ACKNOWLEDGEMENT 4

LIST OF TABLES 6

LIST OF FIGURES 7

LIST OF ABBREVIATIONS 8

LIST OF APPENDICES 9

1. INTRODUCTION 10

1.1 Background and Context 10
1.2 Literature Review 10
1.3 Research Gap 11

2. OBJECTIVE 12

2.1 Main Objective 12
2.2 Specific Objectives 12

3. METHODOLOGY 13

3.1 Phase 1: Data Collection 13
3.2 Phase 2: Scenario Definition 13
3.3 Phase 3: Modeling and Forecasting 13
3.4 Phase 4: Decision Support Engine 13
3.5 Phase 5: Risk Analysis and Sensitivity 13
3.6 Phase 6: User Interface and Reporting 13
3.7 Phase 7: Validation Strategy 13
3.8 Technology Stack 14

HIGH-LEVEL SYSTEM ARCHITECTURE 15

USER REQUIREMENTS 17

5.1 Stakeholder Identification 17
5.2 functional requirements 17
5.3 Non-functional requirement 18

6. COMMERCIALIZATION PLAN 19

BUDGET AND JUSTIFICATION 20

WORK BREAKDOWN STRUCTURE (WBS) 22

GAINT CHART 23

10.REFERENCES LIST 24

11.APPENDICES 26

CALENDAR & EVENT DATA 26

LIST OF TABLES

Table 1: Technology stack summary

Table 2: Functional requirements

Table 3: Non-functional requirements

Table 4: Budget and cost justification

Table 5: Work breakdown structure

List of figures

Figure 1: system architecture diagram

Figure 2: component architecture diagram

List of abbreviations

Abbreviations	Description
SADS	Scenario Analysis & Decision Support
DSS	Decision Support System
CEB	Ceylon Electricity Board
ML	Machine Learning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive Explanations
LIME	Local Interpretable Model-Agnostic Explanations
CSV	Comma-Separated Values
API	Application Programming Interface
GUI	Graphical User Interface
IOT	Internet of Things
ARIMA	AutoRegressive Integrated Moving Average
SARIMA	Seasonal AutoRegressive Integrated Moving Average
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit

LIST OF APPENDICES

Appendix - A: Summary of Required Dataset

1. INTRODUCTION

1.1 Background and Context

The increasing integration of renewable energy sources, changing consumption patterns, and environmental uncertainties have made it more difficult to predict electricity demand accurately. Optimizing energy planning, reducing operating costs, and guaranteeing a steady supply of electricity all depend on accurate demand forecasting. Inaccurate demand forecasting can have a direct effect on utilities, legislators, and communities by causing power shortages, inefficiencies, and needless infrastructure investments [1]–[7].

The nonlinearities and temporal dynamics present in contemporary power systems are frequently difficult for conventional statistical and rule-based forecasting techniques to capture [6], [7]. On the other hand, deep learning architectures like LSTM and GRU, as well as Artificial Intelligence (AI) and Machine Learning (ML) techniques, have shown significant gains in predictive accuracy [6], [10]. Despite these developments, the majority of current methods do not incorporate scenario analysis, which restricts their capacity to assist in making decisions in the face of uncertainty [1], [2], and [4].

In order to quantify the impact of uncertainties on electricity demand and system operations, stakeholders can assess multiple plausible futures through scenario analysis [1], [5]. Scenario analysis offers practical insights for operational planning, strategic decision-making, and policy formulation when combined with Decision Support Systems (DSS). For example, planners can identify peak load periods, predict demand fluctuations, and reduce potential risks by combining AI-based forecasting with interactive scenario simulations [2], [3], [7].

This study suggests a Scenario Analysis and Decision Support System (SADS) that combines interactive scenario modeling with AI-driven forecasting. The system enables energy planners to assess possible outcomes, simulate various demand scenarios, and make evidence-based decisions in the face of uncertainty. By doing this, it supports Sustainable Development Goal (SDG) 7: Affordable and Clean Energy and advances smart energy systems.

1.2 Literature Review

Recent studies underscore the significance of integrating forecasting, scenario analysis, and decision support in modern power systems [8]–[12]:

- Scenario analysis methodologies: IEEE working groups provide standardized frameworks, tools, and best practices for scenario-based electricity planning, enabling rigorous evaluation of system uncertainties [8], [9].
- Decision support frameworks: Contemporary research demonstrates how AI-based forecasting models can be operationalized within decision support platforms to enhance both tactical and strategic energy management [11], [12].
- AI-based electricity demand forecasting: Hybrid and deep learning approaches, including LSTM and GRU models, have been shown to significantly improve prediction accuracy by capturing complex temporal dependencies and nonlinear consumption patterns [10], [11], [12].

Despite these advancements, existing solutions predominantly focus on forecasting accuracy and often neglect the integration of interactive scenario evaluation. There remains a critical need for

comprehensive systems that combine precise demand prediction with practical, actionable decision support, thereby enabling stakeholders to proactively manage risks and optimize electricity planning [8]–[12]

1.3 Research Gap

Current electricity demand forecasting systems largely focus on improving prediction accuracy using statistical methods, machine learning models, and hybrid approaches such as LSTM, GRU, ARIMA, and SARIMA [6], [7], [10]–[12]. While these methods have advanced the ability to generate reliable short-term and long-term forecasts, they do not fully address the uncertainties and dynamic conditions of modern power systems, such as rapid integration of renewable energy, changing consumer behavior, and extreme weather events [1], [5].

Most existing solutions lack interactive scenario analysis, which is critical for exploring multiple plausible futures and evaluating their impact on electricity demand [1], [2], [4]. Without this, decision-makers cannot easily simulate "what-if" situations, such as heatwaves, festivals, industrial demand spikes, or heavy rainfall, limiting their ability to plan operational strategies or mitigate risks [3], [7].

Another significant limitation is the absence of explainable AI (XAI) integration in forecasting models [8], [9]. While AI and machine learning improve predictive accuracy, many models operate as "black boxes," making it difficult for planners and policymakers to understand the key factors driving demand predictions, reducing trust and actionable use of the outputs [10], [11].

Finally, current research rarely combines forecasting, scenario simulation, and decision support into a unified system. Most studies either focus on improving forecast algorithms [6], [10], or on theoretical scenario analysis frameworks [1], [5], but do not integrate these components in a practical tool for real-time energy management [2], [12].

Therefore, there is a clear need for a comprehensive electricity demand management system that:

1. Integrates AI/ML-based forecasting with high predictive accuracy,
2. Provides explainable insights to increase stakeholder trust,
3. Allows interactive scenario-based simulations to test "what-if" situations, and
4. Supports data-driven operational and strategic decision-making under uncertainty [1]–[12].

This integrated approach will bridge the gap between predictive analytics and actionable decision support, making it highly relevant to modern energy planning, risk management, and sustainable electricity supply management.

2. OBJECTIVE

2.1 Main Objective

This project will develop a Scenario Analysis and Decision Support tool that assists planners/utilities/policymakers with analyzing various “what-if” scenarios related to electricity demand management. It will provide the skills needed to model risk prediction using AI and build interactive dashboards for each of the three stakeholders.

2.2 Specific Objectives

1. Use current and previous electricity consumption data along with predictions generated from the weather, calendar/seasonal events, etc. to create realistic prediction scenarios.

2. Create a scenario modeling application capable of modeling situations such as peak electricity demand periods, severe weather conditions, and previously unanticipated issues.

3. Develop an analysis system designed to evaluate potential points of risk and/or impact (both operational and financial) on the electric system.

4. Develop a decision-making support tool that can provide actionable recommendations (i.e., load balancing, contingency planning, and resource allocation) via a user-friendly dashboard.

5. Develop a secure Node.js/Express.js environment with REST APIs for scenario execution; forecasting; logging; and authenticated communications that will enable seamless integration of other components of the platform.

6. Conduct testing and validation to ensure that the module provides accurate and timely results that will allow for effective decision making.

3. METHODOLOGY

This research uses a Design Science Research (DSR) framework to develop, create and evaluate a decision support and scenario analysis module for energy planning, which delivers reliable predictions and insight that facilitate energy policy and operational decisions. Quantitative analysis and modeling techniques were employed as tools to assess the accuracy, usability, and effectiveness of the model.

3.1 Phase 1: Data Collection

Relevant utility sources (the utility companies) will supply information concerning the Demand, Supply, and Operational history involving their respective utilities. Collected Utility Data will provide the foundation for preparing Historical Data and Weather Data. The latest Market Data will also be incorporated into the scenario analysis process as needed. The above mentioned datasets will be subjected to quality checks and pre-process, prior to using them in formulating scenarios.

3.2 Phase 2: Scenario Definition

Energy planning scenario design is determined by many factors such as renewable integration, demand variability, extreme weather behavior, and other policy issues. These scenarios provide plausible futures for making decisions and evaluating risk.

3.3 Phase 3: Modeling and Forecasting

Electricity consumption is predicted using AI/ML models that include a variety of regression, neural net, and hybrid forecasting models. Each model is built for particular scenarios through the application of historical data, validation testing, and optimization for accuracy and reliability.

3.4 Phase 4: Decision Support Engine

In order to assist in developing strategies for successful action, we must have a decision support engine that will provide useful and actionable insights by analyzing scenario outcome data in real-time based on specific criteria. Examples of metrics that will be analyzed include predicted demand, peak load risk, cost implications and the ability to integrate renewable resources.Results obtained through the analysis of each metric will be displayed in easily readable formats via visual dashboards designed specifically for energy planners and decision-makers alike.

3.5 Phase 5: Risk Analysis and Sensitivity

The evaluation of uncertainty and sensitivity in scenario outputs is critical to the probabilistic risk analysis procedure where potential high risk outcomes are identified as well as key variables affecting the stability of the energy system. This allows stakeholders to develop and implement appropriate actions to mitigate these risks.

3.6 Phase 6: User Interface and Reporting

A simple user interface enables users to choose scenarios, access results, and generate reports. Graphical charts and tables, as well as alerts, help decision-makers interpret predictions and make educated decisions.

3.7 Phase 7: Validation Strategy

Expert review by Energy Planners and historical back-testing are used to validate model predictions and evaluate the accuracy, usability, and relevance of the outputs for the scenarios. The assessments are used to guide users in improving model accuracy, usability, and interface design.

3.8 Technology Stack

Category	Technology	Justification
Data Analysis & Modeling	Python (Pandas, NumPy)	Widely used for data manipulation and statistical analysis; easy to integrate with ML models
Data Analysis & Modeling	TensorFlow / Scikit-learn	Supports machine learning models for energy demand forecasting and scenario simulations
Simulation & Visualization	Matplotlib / Plotly / Dash	interactive charts and dashboards for scenario results; easy for decision-makers to interpret
Backend / API	Node.js / Express.js	Handles scenario requests, serves model results to the frontend; fast JSON handling
Backend Hosting	Firebase Cloud Functions / AWS Lambda	Serverless auto-scaling; no server management; integrates natively with Firebase ecosystem
Database	Firebase	Stores historical energy data, scenario results; supports real-time access and secure storage
Authentication	Firebase Auth + JWT	Secure access control for analysts and planners
Frontend / Dashboard	Collaborative development, automated testing, and deployment pipeline)	Interactive dashboards for visualizing scenario analysis and KPIs
Version Control / Cl	GitHub + GitHub Actions	Collaborative development, automated testing, and deployment pipeline

4. HIGH-LEVEL SYSTEM ARCHITECTURE

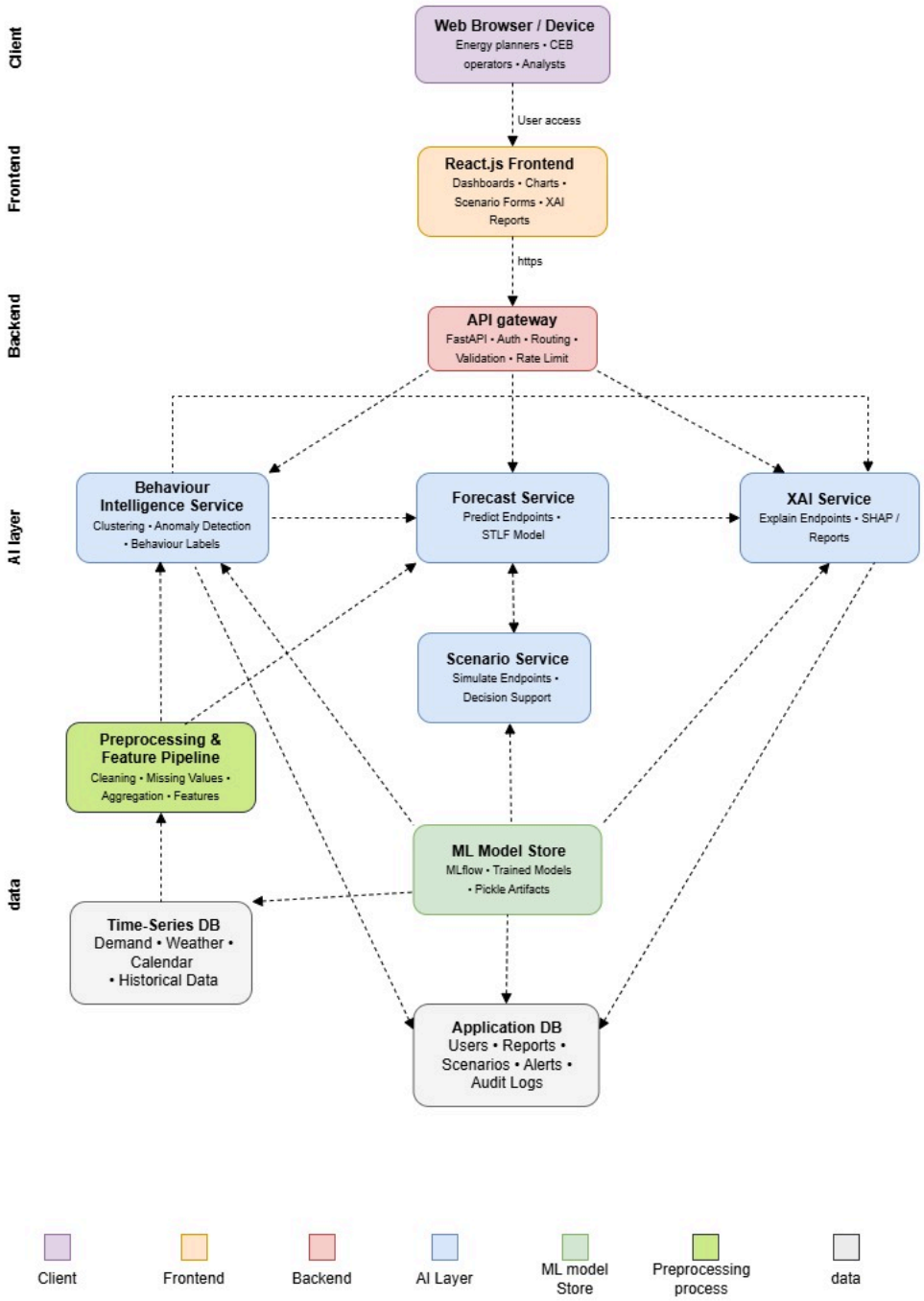
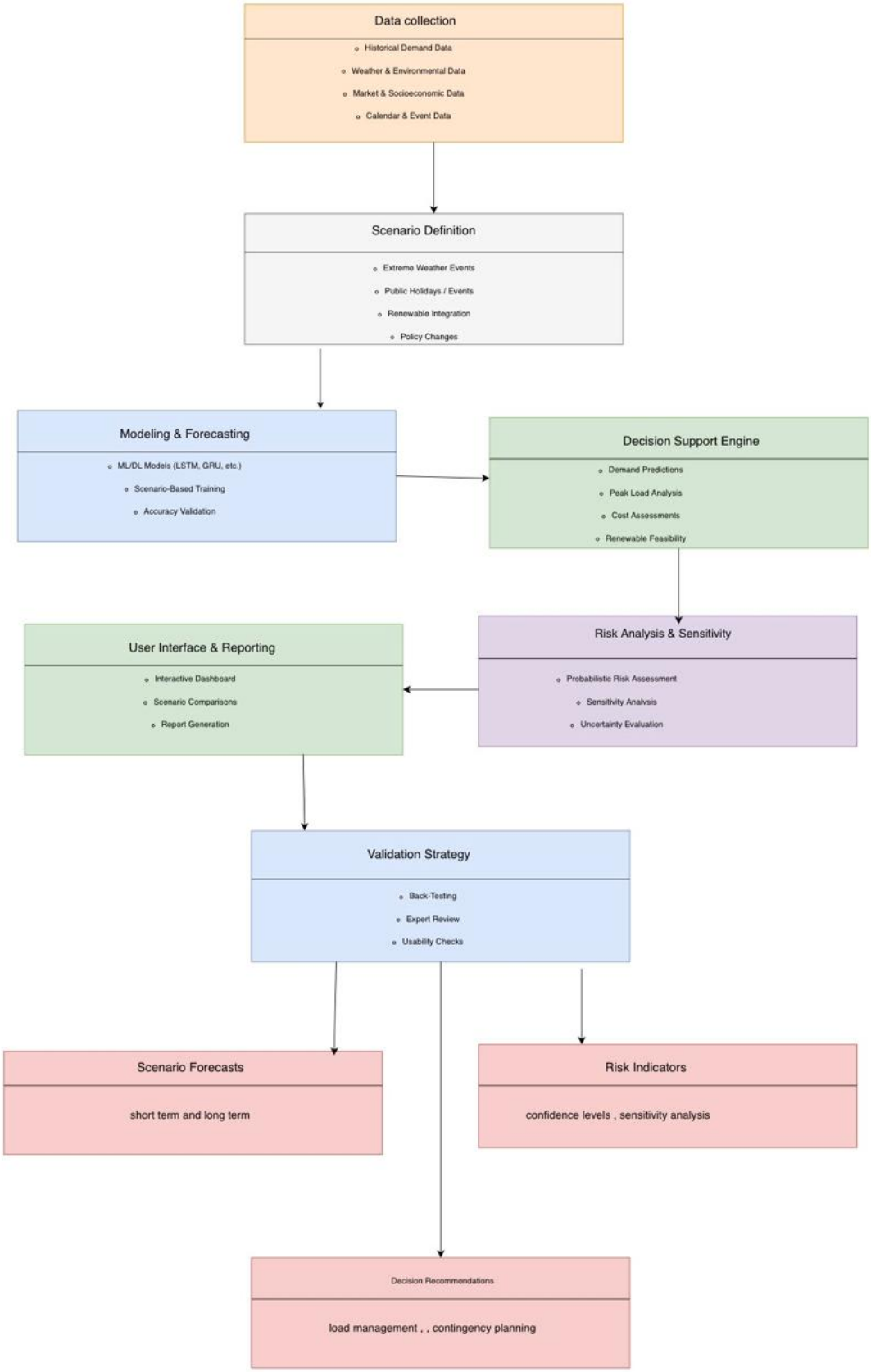


Figure 1: system architecture diagram



2: component architecture diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Main Use: Planners and analysts working with energy resources, creating scenarios, and providing information for decision.

Secondary Users: Staff at utility companies and government entities that are looking for feedback from the energy analyst.

Tertiary Stakeholders: Researchers and/or consultants looking for information about energy and energy planning.

Systems Administrators: Your research team who will be responsible for maintaining the systems and dashboards.

5.2 functional requirements

FR	Category	Requirement
FR1	Data Input	System shall accept historical electricity demand, weather, and event data for simulation
FR2	Scenario Setup	Users shall be able to define multiple “what-if” scenarios (e.g., high demand, weather changes, public events)
FR3	Forecast Engine	System shall use AI/ML models to predict electricity demand under different scenarios.
FR4	Decision Support	System shall provide recommendations for grid management, load balancing, or emergency actions.
FR5	Visualization	System shall display scenario outputs with charts, graphs, and tables for easy analysis.
FR6	Confidence & Risk	System shall provide confidence scores and risk indicators for each forecasted scenario
FR7	Report Generation	Users shall be able to export scenario results as PDF/Excel for planning or documentation.
FR8	Real-time Updates	System shall update simulations in real time when input data or scenario parameters change.
FR9	Backend API	Node.js or cloud backend shall handle data processing, model inference, and storage securely.
FR10	Offline Mode	Web or desktop UI shall allow easy scenario selection, parameter adjustment, and visualization

5.3 Non-functional requirement

Category	Requirement
Performance	The system should run scenario simulations and provide decision support quickly. Small datasets should process in under 100ms, and dashboard updates should show within 200ms.
Accuracy	Forecasts and scenario outputs should be reliable, with at least 90% prediction accuracy and minimal false recommendations
Usability	Users should be able to use the system without training. Dashboard feedback and results should appear within 1 second after running a scenario.
Reliability	The system should be stable and available 99.9% of the time. If part of the system fails, it should fallback to simpler models to keep running.
Privacy	All data must be secure. Sensitive information should not be shared externally, and logs should be anonymized
Compatibility	The system should work on web browsers and standard devices used by electricity planners. Cloud services should support Windows and Linux servers.
Security	All communication should be encrypted with TLS. User access must be controlled by roles, and admin functions must have strict authorization.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Electric utility companies and electricity grid managers are the primary target customers for the Scenario Analysis & decision Support System in Sri Lanka. These companies require tools to effectively forecast electric demand and plan for future developments. Municipal planners, emergency management agencies, and renewable energy companies will also leverage the predictive intelligence contained in Scenario Analysis to make informed decisions about operations. There is also potential for utility consultants and smart grid technology manufacturers to utilize the Scenario Analysis System. An example of a typical customer is an electricity grid manager who needs to be able to view various possible future demand scenarios very quickly so that he/she can make thoughtful decisions.

6.2 Value Proposition

Users can see what their electricity requirements will be based on different scenarios (for example, extreme heat, heavy rain, or public events) using this system. In addition to being user-friendly, the system incorporates forecasting, risk assessment, and decision-making aides into one comprehensive platform. Users can combine multiple data sets to create simple visualizations and make smart decisions about energy use based on the outcome of those visualizations.

6.3 Revenue Model

Utilities with a need for continued access to scenario simulation; purchase annual license, consulting services - set multiple scenarios/ customize systems; also potentially offer 3rd parties an API with a usage- based pricing model to integrate into other energy management tools.

6.4 Competitive Advantage

This system has an original combination of scenario analysis, forecasting and decision support systems in terms of the Sri Lanka electric utility framework. It must consider local weather patterns, new loading levels from various sources including renewables, and existing operating limits or constraints. Therefore, unlike common general audience forecasting tools, the output from this system will be based on the instantaneous application of data to provide actionable results/ recommendations at any time via an on-premise or cloud system.

6.5 Intellectual Property Considerations.

All algorithm methods used for scenario modeling, risk analysis, and dashboard design are privately owned. This computer system will utilize both commercial (paid) license and openaccess (academic research / instructional) license methods to be accessed from agencies of all types.

7. BUDGET AND JUSTIFICATION

Item	Qty	Unit (LKR)	Total (LKR)	Justification
Laptop/desktop (16GB 1 RAM , i7)	1	150,000	150,000	Primary development and testing device for Python ML modeling and Node.js backend
external Hard Drive (2TB	1	12,000	12,000	Storage of historical electricity demand, weather data, and scenario outputs
Google Colab Pro+ (12 months)	12	4,500	54,000	GPU-accelerated AI model training (LSTM, GRU, regression models)
Firebase Blaze Plan (12 months)	12	3,000	36,000	Cloud database for scenario storage, real-time access, and backend hosting
Data Acquisition Fees	1	20,000	20,000	Purchase or access to proprietary electricity demand and weather datasets
transport & Site Visits	3	6000	18,000	Travel to utility offices, meteorological departments, or municipal sites for data collection and meetings
Visualization & Dashboard Tools (Plotly/Dash/Power BI)	1	15,000	15,000	Interactive dashboards for scenario outputs and decision support
Miscellaneous (printing, stationery, minor software)	1	5,000	5000	Documentation, reports, and project presentation materials
TOTAL BUDGET			LKR 310,000	

A laptop is needed to create a machine learning model and a scenario simulation system. To expedite the training of AI models such as LSTM a nd GRU via GPU resources, we will use Google Colab Pro+. Firebase Blaze plan will host the database and the node.js backend for storing data and running scenario simulations. The cost of transport will cover meetings, collecting data, and consulting with key stakeholders. We will use visualization to develop dashboards and graphs to allow planner's to visualise the scenario results. Miscellaneous cost will cover printing and reporting and other small research related costs.

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS	Task	Sub-Tasks	Timeline
1.0	Research & Requirements analysis	Literature review on scenario analysis and decision support systems, identify key variables affecting electricity demand (temperature, rainfall, events), define system requirements	Jan – Feb 2026
2.0	Data Collection & Preparation	Collect historical electricity demand data, collect weather/environmental data, preprocess, and clean datasets	Feb – Mar 2026
3.0	Forecast Data Integration	integrate electricity demand forecasts from AI prediction component, validate data inputs for scenario simulation	Feb – Mar 2026
4.0	Scenario Modeling	Define possible future scenarios (temperature rise, heavy rainfall, public events), develop scenario simulation logic, test different scenario conditions	Feb – Mar 2026
5.0	Decision Support Development	Design decision rules and recommendations, implement risk assessment model, generate suggested actions for energy management	Mar – Apr 2026
6.0	Visualization & User Interface	Develop dashboard for scenario inputs, implement graphs and charts for demand predictions, create scenario comparison interface	Apr – May 2026
7.0	System Integration	Connect scenario module with forecasting and main system components, ensure proper data flow between modules	Apr – May 2026
8.0	Testing & Validation	Test scenario simulations, verify accuracy of recommendations, perform usability testing	May – Jun 2026
9.0	Performance Optimization	improve simulation efficiency, optimize data processing and system performance	May – Jun 2026
10.0	Documentation & Final Submission	Prepare system documentation, write component report, finalize proposal and project submission	Jun – Aug 2026

9. GAINT CHART

System Development Project Timeline – 2026													
		● Semester 1 (Jan – Jun 2026)						● Semester 2 (Jul – Dec 2026)					
Phase	Activity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Phase 1 – Research & Design													
Phase 1	1.0 Research Requirements	▪	▪										
Phase 1	2.0 System Design		▪	▪									
Phase 2 – Data & Model													
Phase 2	3.0 Dataset Preparation			▪	▪	▪							
Phase 2	4.0 Model Development					▪	▪	▪					
Phase 3 – Implementation													
Phase 3	5.0 Backend Development							▪	▪	▪			
Phase 3	6.0 Frontend Development									▪	▪		
Phase 3	7.0 System Integration											▪	
Phase 4 – Testing & Delivery													
Phase 4	8.0 Testing & Evaluation												▪
Phase 4	9.0 Documentation & Final												▪
Summary	Tasks per Semester	Semester 1: Tasks 1.0 – 4.0 | Research → Model Development						Semester 2: Tasks 4.0 – 9.0 | Development → Delivery					

10.References list

[1] J. R. Santos, M. D. de Andrade, and J. P. Lopes, "A review of scenario analysis methods in planning and operation of modern power systems: Methodologies, applications, and challenges," *Electric Power Systems Research*, vol. 205, 107722, 2022. [Online]. Available: https://doi.org/10.1016/j.epsr.2021.107722. [Accessed: 13-Mar-2026]. :contentReference[oaicite:0]{index=0}

[2] "A decision support framework for electric demand planning in distribution systems during extreme minimum temperatures," *Energy Reports*, vol. 14, pp. 1137–1148, 2025. [Online]. Available: https://doi.org/10.1016/j.egyr.2025.06.020. [Accessed: 13-Mar-2026]. :contentReference[oaicite:2]{index=2}

[3] A. Jozi, T. Pinto, I. Praça, and Z. Vale, "Decision support application for energy consumption forecasting," *Applied Sciences*, vol. 9, no. 4, p. 699, 2019. [Online]. Available: https://www.mdpi.com/2076-3417/9/4/699. [Accessed: 13-Mar-2026]. :contentReference[oaicite:4]{index=4}

[4] "Scenario planning for electricity supply," *Energy Policy*, vol. 31, pp. 1647–1659, Dec. 2003. [Online]. Available: https://doi.org/10.1016/S0301-4215(02)00231-8 [Accessed: 13-Mar-2026]. :contentReference[oaicite:6]{index=6}

[5] [5] M. Kanté, Y. Li, and S. Deng, "Scenarios analysis on electric power planning based on multi-scale forecast: A case study of Taoussa, Mali from 2020 to 2035," *Energies*, vol. 14, no. 24, p. 8515, 2021. [Online]. Available: https://doi.org/10.3390/en14248515. [Accessed: 13-Mar-2026]. :contentReference[oaicite:8]{index=8}

[6] S. H. Z. Rashed and W. N. M. Zaki, "Machine-learning-based electric power forecasting: A review," Sustainability, vol. 15, no. 14, Article 11299, 2023. [Online]. Available: https://www.mdpi.com/2071-1050/15/14/11299?utm_source=chatgpt.com. [Accessed: 13-Mar-2026].

[7] S. A. Hasnain et al., "Short-term electricity load forecasting using hybrid intelligent models," Sustainability, vol. 15, no. 16, Article 12311, 2023. [Online]. Available: https://www.mdpi.com/2071-1050/15/16/12311?utm_source=chatgpt.com. [Accessed: 13-Mar-2026].

[8] IEEE Power & Energy Society, "Working Group on Energy Forecasting and Analytics," IEEE PES Subcommittee on Power System Economics. [Online]. Available: https://site.ieee.org/pes-psope/subcommittees/power-system-economics-subcommittee/working-group-on-energy-forecasting-and-analytics/?utm_source=chatgpt.com. [Accessed: 13-Mar-2026].

[9] IEEE Power & Energy Society, "Open Access Journal of Power and Energy Systems – Energy Forecasting Competition Insights," IEEE Resource Center, Dec. 2021. [Online]. Available: https://resourcecenter.ieee.org/publications/oajpe-open-access-papers/pes_tp_oajpe-00101-2021_12-22?utm_source=chatgpt.com. [Accessed: 13-Mar-2026].

[10] M. Saglam, C. Spataru, and O. A. Karaman, "Electricity demand forecasting with use of artificial intelligence: The case of Gokceada Island," Energies, vol. 15, no. 16, Article 5950, 2022. [Online]. Available: https://www.mdpi.com/1996-1073/15/16/5950?utm_source=chatgpt.com. [Accessed: 13-Mar-2026].

[11] M. Farrokhabadi, J. Browell, Y. Wang, S. Makonin, W. Su, and H. Zareipour, “Day-Ahead Electricity Demand Forecasting Competition: Post-COVID Paradigm,” *IEEE Power & Energy Society Open Access Journal of Power and Energy Systems*, vol. 9, pp. 1–7, 2022. [Online]. Available: https://resourcecenter.ieee.org/publications/oajpe-open-access-papers/pes_tp_oajpe-00101-2021_12-22?utm_source=chatgpt.com. [Accessed: 13-Mar-2026].

[12] T. Hong, P. Pinson, Y. Wang, R. Weron, D. Yang, and H. Zareipour, “Energy Forecasting: A Review and Outlook,” *IEEE Power & Energy Society Open Access Journal of Power and Energy Systems*, vol. 7, pp. 1–13, Oct. 2020. [Online]. Available: https://resourcecenter.ieee.org/publications/oajpe-open-access-papers/pes_tp_oajpe-00054-2020_12-20?utm_source=chatgpt.com. [Accessed: 13-Mar-2026].

11.APPENDICES

1. Temporal Data

    - Timestamp — recorded at 15-minute intervals, serving as the primary time index
    - Hour of Day — numeric representation of the hour (0–23)
    - Day of Week — identifier for weekly cycles (0 = Monday, 6 = Sunday)
    - Month — month of the year (1–12)
    - Season — seasonal classification (Summer, Winter, Monsoon, etc.)

2. Target Variable
    - Electrical Load Demand (kW) — the main forecasting variable, representing consumption patterns to be predicted by CNN-LSTM and Transformer models

3. Weather & Environmental Data
    - Temperature (°C) — average temperature per interval
    - Humidity (%) — relative humidity levels
    - Wind Speed (m/s) — average wind speed
    - Rainfall (mm) — precipitation levels
    - Solar Irradiance (W/m²) — sunlight intensity affecting demand

4. Socioeconomic & Pricing Data
    - GDP (USD) — indicator of economic activity influencing demand
    - Per Capita Energy Use (kWh) — average consumption per person
    - Electricity Tariff (LKR/kWh) — prevailing electricity price in Sri Lankan Rupees

Calendar & Event Data

- Public Holidays / Events — binary flag (1 = event/holiday, 0 = none)
- Special Occasions — optional categorical variable for demand spikes (e.g., festivals, sporting events)